Practical No. 7 Competitive Code

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Code:

```
import java.util.ArrayList;
import java.util.Arrays;

class Solution {

  boolean[] visited;

  boolean check(int N, int M, ArrayList<ArrayList<Integer>> Edges) {

    ArrayList<ArrayList<Integer>> adj = new ArrayList<>();
    for (int i = 0; i <= N; i++) {
      adj.add(new ArrayList<>());
    }

    for (ArrayList<Integer> edge : Edges) {
      int u = edge.get(0);
      int v = edge.get(1);
      adj.get(u).add(v);
      adj.get(v).add(u);
    }

    this.visited = new boolean[N + 1];

    for (int i = 1; i <= N; i++) {
      Arrays.fill(visited, false);

      if (dfs(i, 1, N, adj)) {
        return true;
      }
```

```
    }

    return false;
  }
  boolean dfs(int u, int count, int N, ArrayList<ArrayList<Integer>> adj) {
    visited[u] = true;

    if (count == N) {
      return true;
    }
    for (int v : adj.get(u)) {
      if (!visited[v]) {
        if (dfs(v, count + 1, N, adj)) {
          return true;
        }
      }
    }
    visited[u] = false;
    return false;
  }
}
```

Submission: